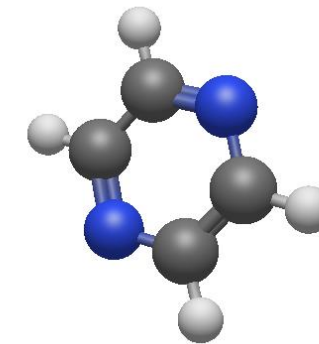
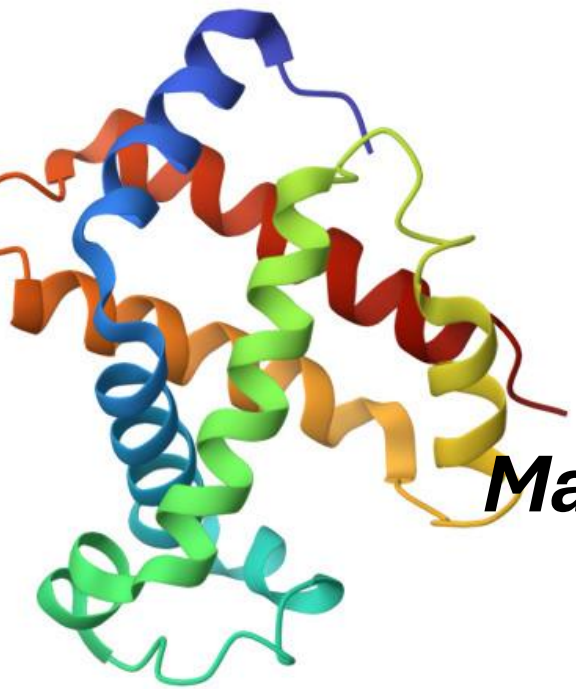


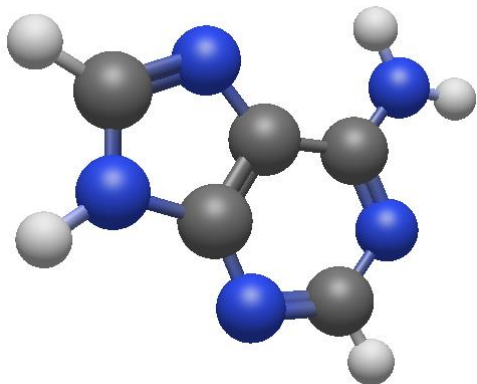


Machine Learning-Enhanced Computational Modeling of Metal-Protein Interactions

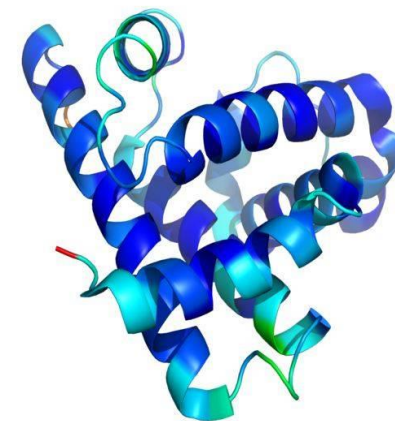
Carter Close, Isabel Isahakian, Yanliel Feliciano

SIMCODES REU

Mentors: Gregory Curtin (Chem), Qi Li (CS), Peng Xu (Chem)



Project background



- Improve the Effective Fragments Potential (EFP) method.
 - Benchmarks EFP using S22, S66, L7 databases
 - Study monomers and dimers interactions
 - Integrate Machine learning
- Validate EFP2 before applying it to complex metal-protein systems
 - Apply the method to solvated metal ions and proteins.

Effective Fragment Potential (EFP)

A cheap way to model non-binded interaction energy between monomer fragments

$$E^{EFP} = \boxed{E^{Coul} + E^{Pol} + E^{Disp}} + E^{ExRep} + E^{CT}$$

These three have damping functions

For example, dispersion energy (E^{Disp}) has three terms

$$E^{Disp} = E(R^{-6}) + E(R^{-7}) + E(R^{-8}) = \sum_{n=6,7,8} E(R^{-n})$$

Each term is of the form damping function * analytic expression

$$E(R^{-n}) = f_n^{disp} * \text{analytic expression}$$

Two types of f_n^{disp} :

Overlap-based

$$f_n^S(i, j) = 1 - S_{ij}^2 \sum_{n=0}^{6 \text{ or } 7 \text{ or } 8} \frac{-2 \ln |S_{ij}|^{n/2}}{n!}$$

Tang-Toennies

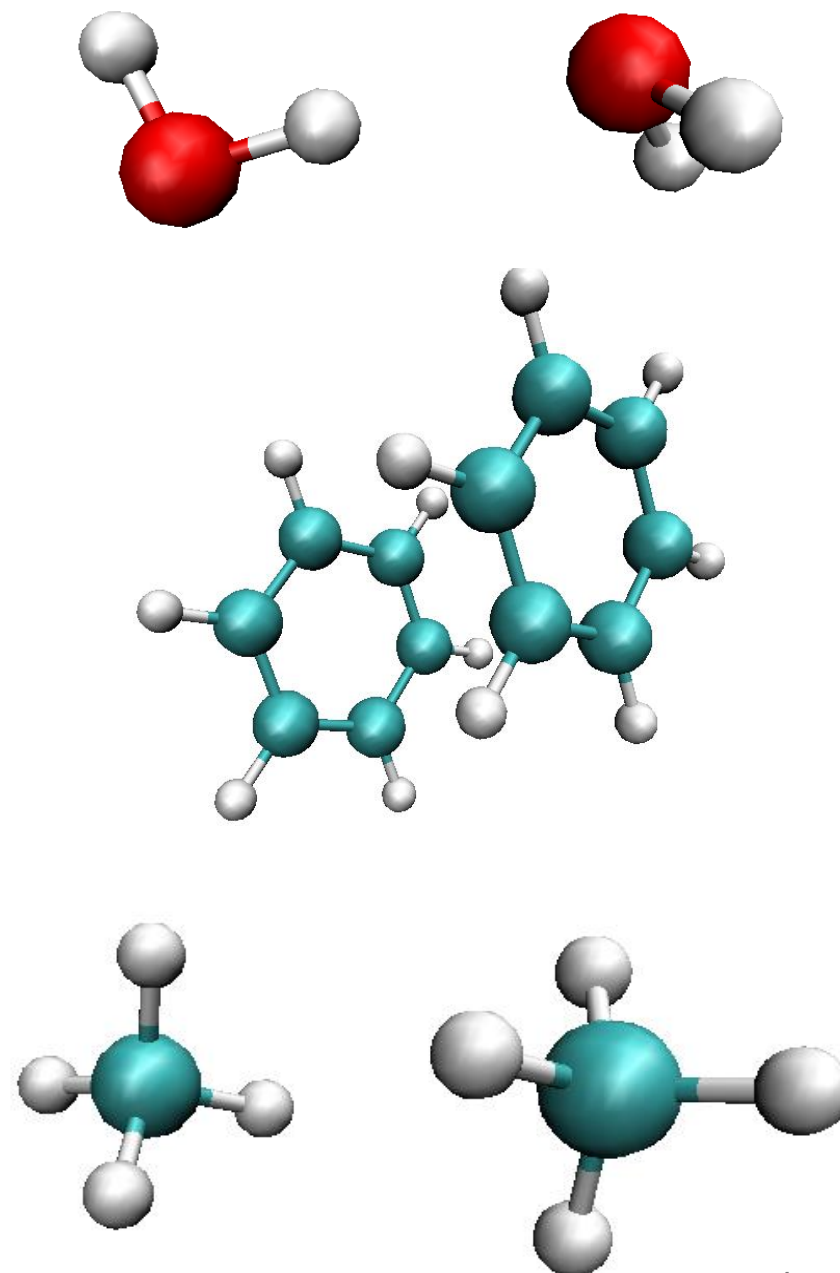
$$f_n^{TT}(i, j) = 1 - \left(\sum_{n=0}^{6 \text{ or } 7 \text{ or } 8} \frac{(bR_{ij})^n}{n!} \right) e^{-bR_{ij}}$$

S_{ij} and R_{ij} are two-body terms and are computed on-the-fly; $b = 1.5$

(1)
Gordon, M. S.; Freitag, M. A.; Bandyopadhyay, P.; Jensen, J. H.; Kairys, V.; Stevens, W. J. The Effective Fragment Potential Method: A QM-Based MM Approach to Modeling Environmental Effects in Chemistry. **2001**, 105 (2), 293–307. <https://doi.org/10.1021/jp002747h>.

S22 Benchmark Set

- Conducting EFPs to validate ground truth needed for ML
- Contains 22 molecular dimers
- Used the benchmark dataset to evaluate the accuracy of EFP
- Provides a strong benchmark, but is limited in size and biological relevance
- Led to the development of S66 and L7



EFP Method using S66 and L7

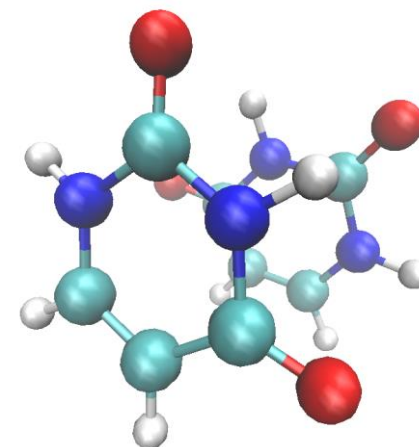
Objective: Validate the accuracy of EFP2 using benchmark database.

- 1) Perform EFP2 calculations
- 2) Analyze the S66 and L7 database
- 3) Compare EFP interactions energy with published values and evaluate different interactions

Why S66 and L7 databases?

S66: Intermolecular forces of hydrogen bonds, dispersion and others. (Dimers)

L7: $\pi - \pi$ stacking and long-range interactions (Complex molecules)



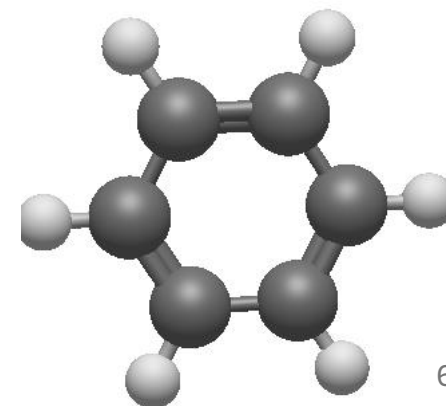
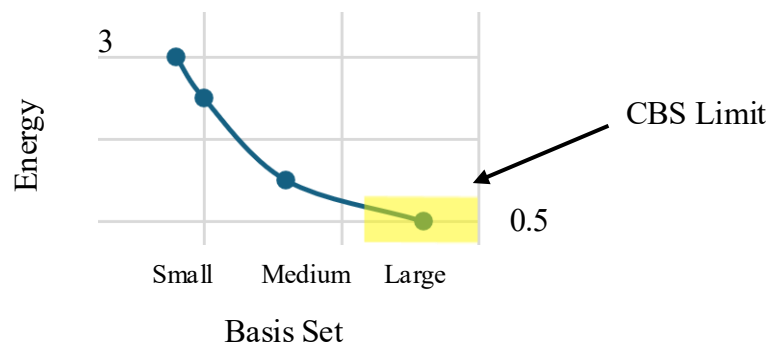
Complete Basis Set (CBS)

- The CBS is obtained by extrapolating calculations to a large basis set.
- Produces interaction energies that approximate the QM solution for different molecules.

Why is important?

- Considered as a reference standard
- Provides ground truth for benchmark databases
- Determine how accurately EFP reproduces results because provides references energies.

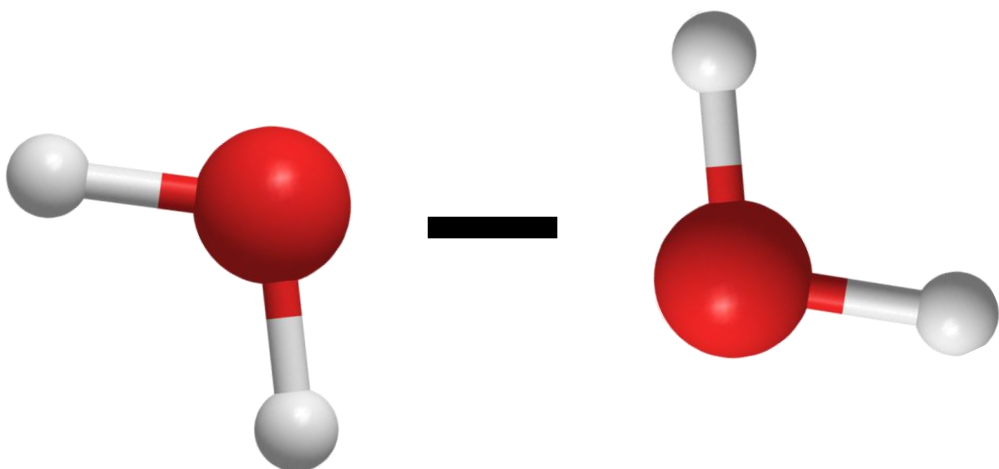
Interaction (Dispersion)	Energy (kcal/mol)
Benzene-Benzene	-2.82



Fragment-Fragment Interaction energies

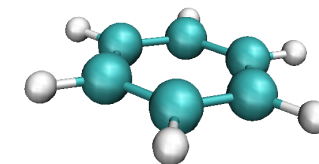
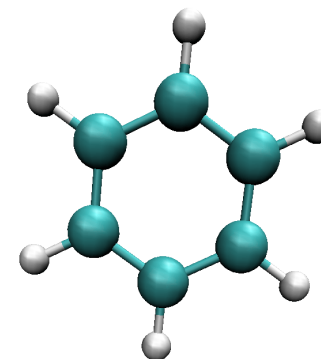
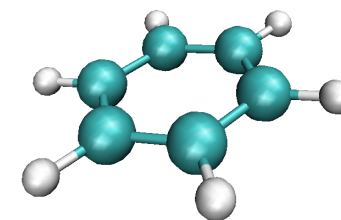
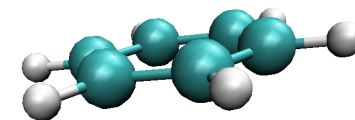
Energy Component	EFP Energy (kcal/mol)
Electrostatic	-7.94
Repulsion	5.43
Polarization	-1.25
Dispersion	-1.16
Charge Transfer	-0.49

Total EFP Energy= -5.41 kcal/mol



Computer Science Applications

- Compile working version of GAMESS on Nova
- GAMESS
 - Build up S22 database
 - Improve input structure and coordinates for EFP2 runs
 - Error handling
- Parser for log file output
 - Obtain usable CSV file from EFP2 log files



Workflow

- (1) Input XYZ coordinates for one monomer → .efp file containing potential energies

- (2) Two or more .efp files for monomers → EFP2 log file containing energies for each part of E^{EFP} → parsed for CSV

```
Between: frag= 1 LM0= 15 and frag= 2 LM0= 15
S(IJ)= -8.086532223960791E-003
R(IJ)= 7.46616692005966
overlap F6= 0.998599428029532
overlap F7= 0.998563403976937
overlap F8= 0.998549426430857
undamped E6= -3.691578370697306E-005
undamped E7= -1.663306292948648E-006
undamped aniso E8= -2.520868845469089E-005
undamped E8= -1.479447089421116E-005
distributed aniso E6= -5.388983861860935E-003
distributed aniso E8= -2.418836308664853E-003
```

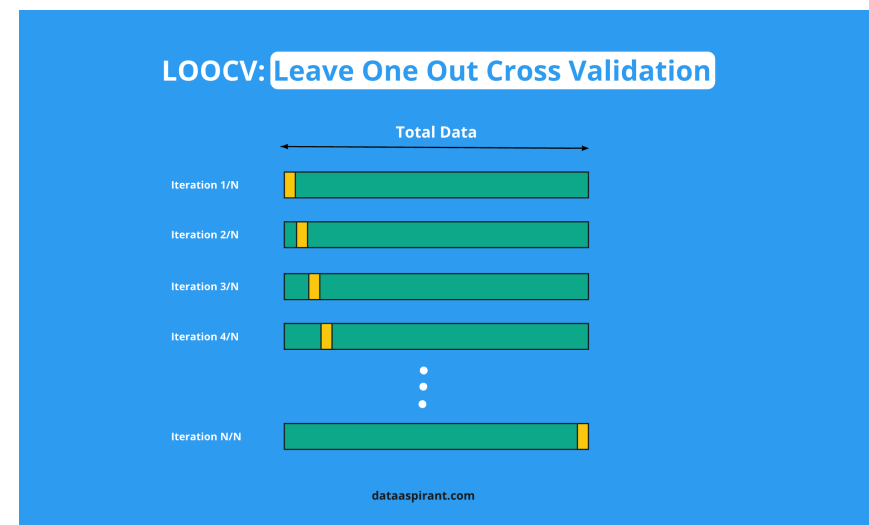
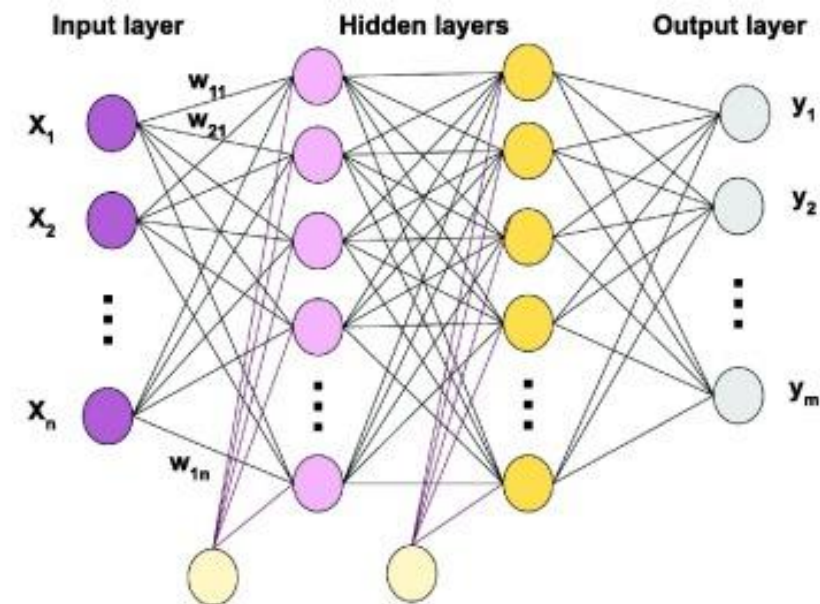
```
===== Parsing cyclo_EFP2.log =====
```

```
Found 4050 interactions.
```

```
{'frag1': 1, 'LM01': 1, 'frag2': 2, 'LM02': 1, 'S(IJ)': 0.004238982072528212, 'R(IJ)': 8.99039415427784, 'overlap F6': 0.999535237364422, 'overlap F7': 0.999519861942668, 'overlap F8': 0.999513508855357, 'undamped E6': -1.227640672603852e-05, 'undamped E7': -1.065317216105453e-06, 'undamped aniso E8': -5.659871932050316e-06, 'undamped E8': -3.700674140234228e-06}
```

Machine Learning Applications

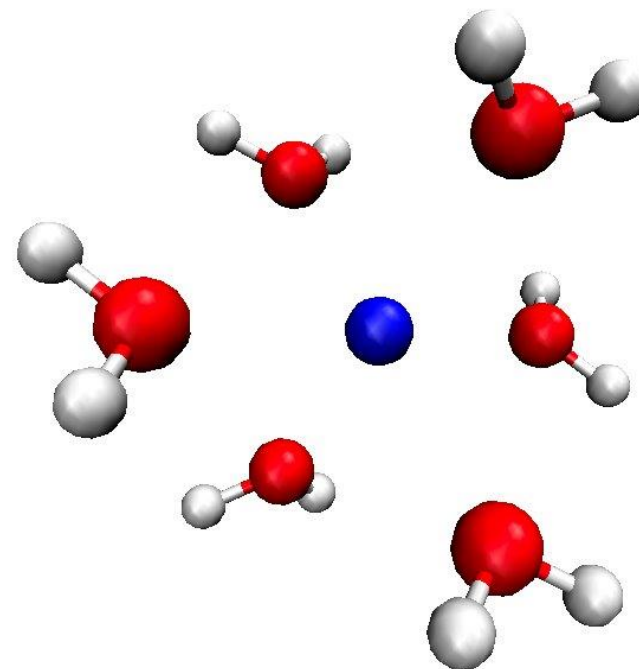
- Goal: Minimize EFP2 energy calculations to ground truth and find generalized functions for damping
- Data: Localized molecular orbitals (LMO), damping data, total energy
- Multi-Layer Perceptron Regressor (MLPRegressor)
 - Implemented in sklearn with integration in fit/predict pipeline
 - Limited to CPU, less customization
- Leave-One-Out Cross Validation (LOOCV)
 - Useful for small datasets (~88 database dimers)
 - Bias/Variance tradeoff – low bias, high variance



Solvated Metals: Background

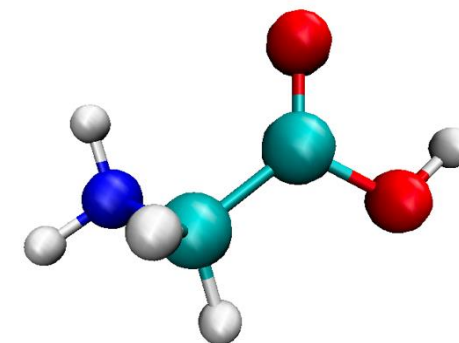
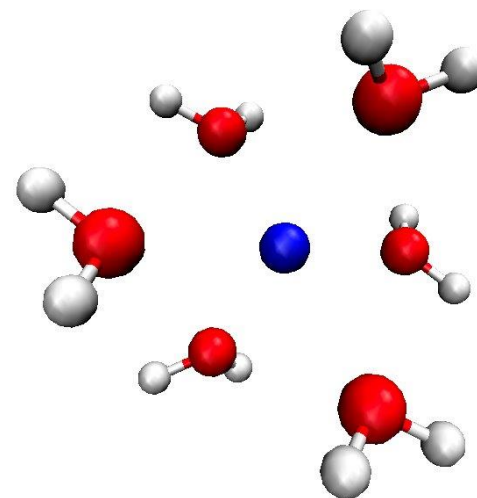
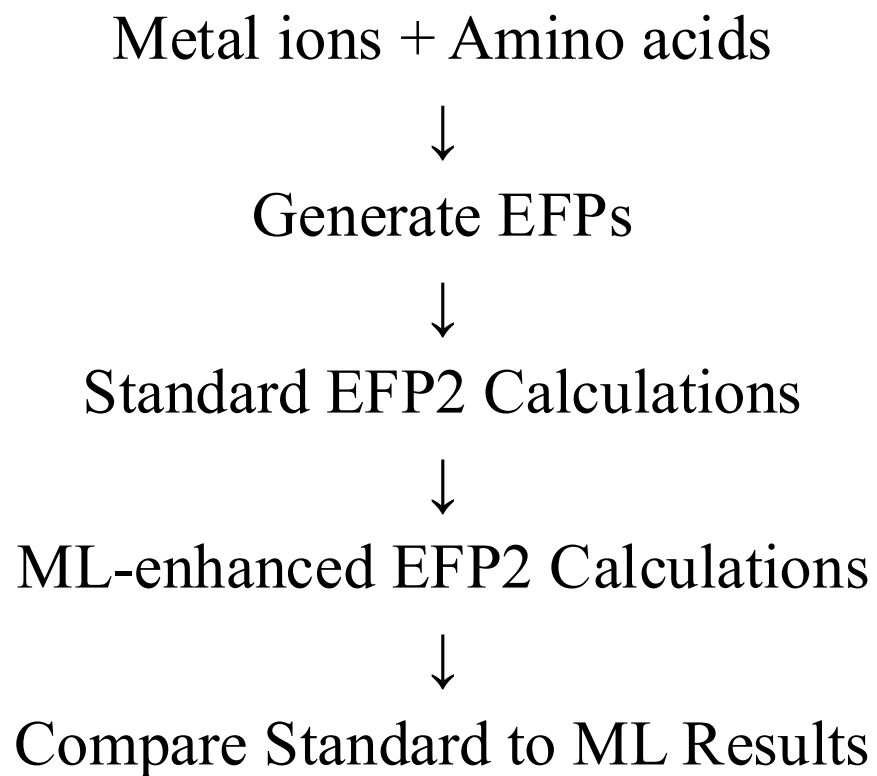
Relevant Background:

- S22, S66, and L7 provide benchmark data
- Biologically relevant metal–amino acid interactions remain underrepresented, no benchmark data
- Benchmark needed to compare standard EFP2 calculations with ML enhanced EFP2 calculations



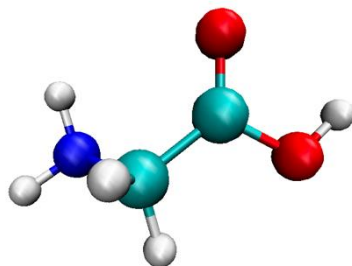
Solvated Metals: Background

Goal:

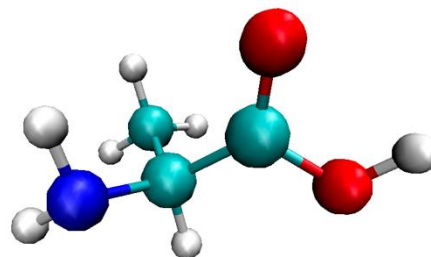


Solvated Metals: Progress

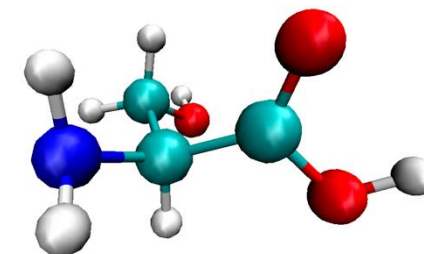
Class	Completed
Alkali metals (Na ⁺ , K ⁺) EFP	2/2 ✓
Alkaline earth metals (Mg ²⁺) EFP	1/1 ✓
Amino acids EFP	6/20
EFP 2 Calculations	0/60



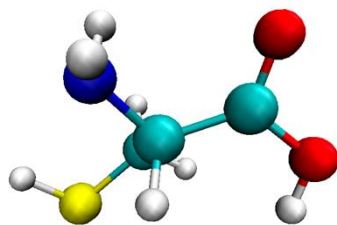
Glycine



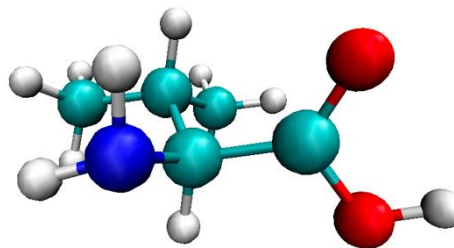
Alanine



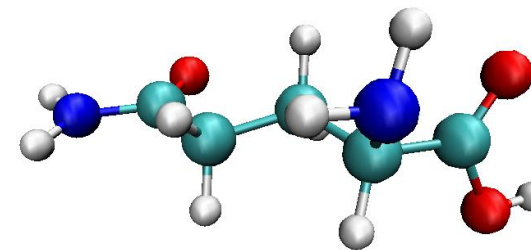
Serine



Cysteine



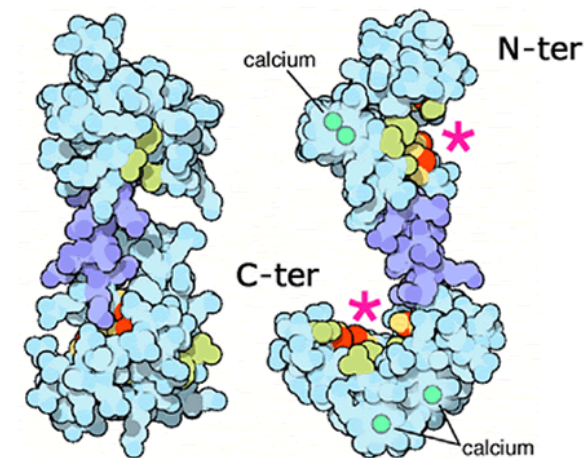
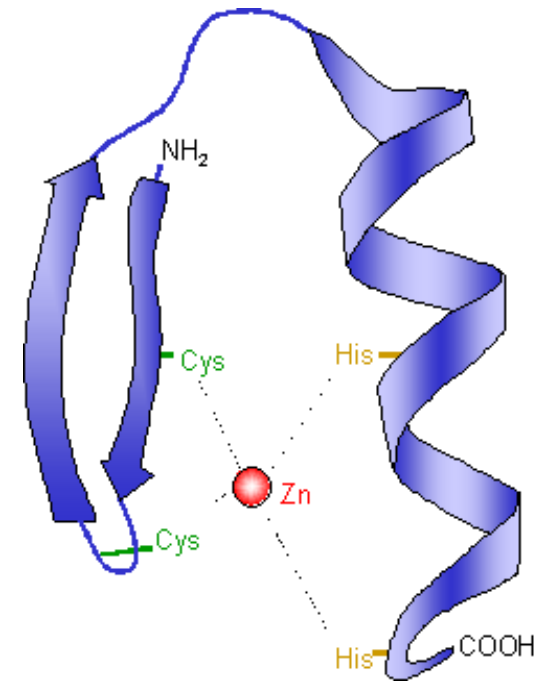
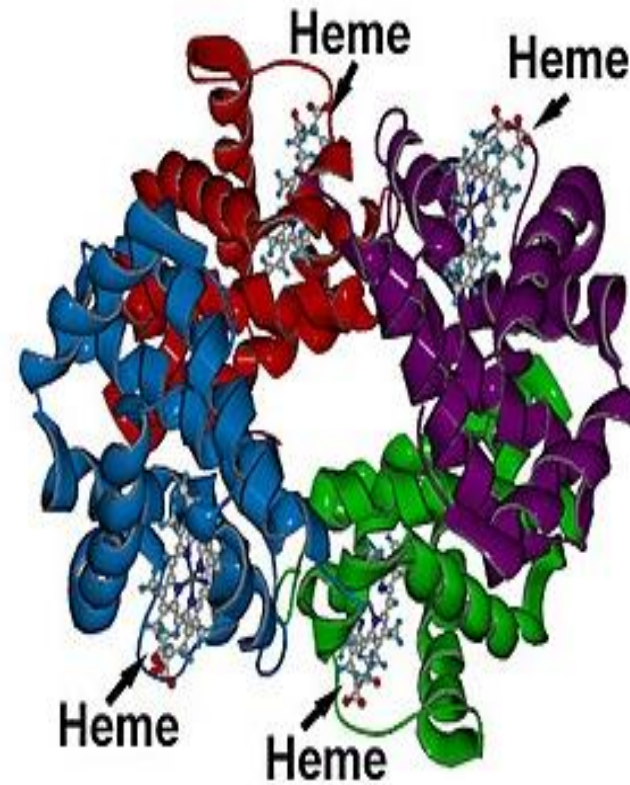
Valine

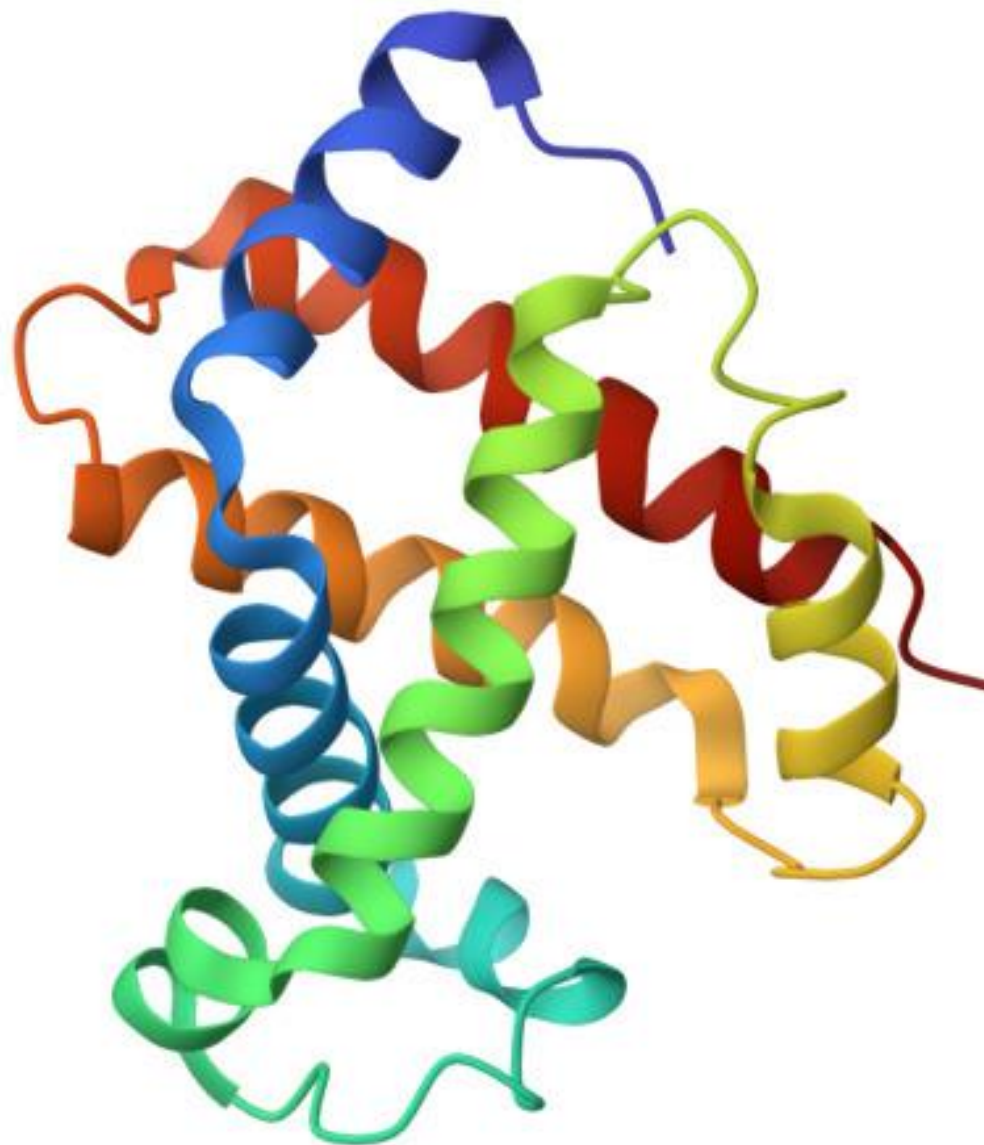


Glutamine

Applications

- Metals are essential for enzyme catalysis and DNA binding
- Zinc finger protein motifs, heme protein, calmodulin
- Na^+ and K^+ : Stabilize protein surfaces
- Mg^{2+} : Nucleic acid binding and enzyme activity





Conclusion

- Build up S22, S66, L7 databases with EFP2 calculations to use for benchmarks and extra data points
- Input data into ML models to minimize towards CBS and optimize parameters
- Generalize to solvated metal systems by comparing base EFP2 calculations to ML-enhanced calculations